REFINING A PRESENTATION TOWARDS CONSTRUCTIBLE SHEAVES ON A SITE WITH ENOUGH QUASI-COMPACT OBJECTS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file sites-modules.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement and the proof contain commutative diagrams typeset with Xy-pic; the retained two-slice preamble of this source supplies the native Xy-pic setup, so they compile as in the source.

Editorial note. The counted span cites a situation and a refinement lemma, both recorded in assembly.external_references with their canonical tags as navigation pointers.

Allowed standard prerequisites (editorial).

- Quasi-compact objects admit finite subcoverings of a basis covering. Stacks Project, Modules on Sites, Lemma lemma-silly-quasi-compact (tag 09B0). Provides the finite subfamilies K'_i and M'_j used at both refinement steps.
- The quasi-compact objects situation and its enough-objects hypotheses. Stacks Project, Modules on Sites, Situation situation-quasi-compact-objects (tag 09AZ). Supplies the ambient hypotheses (items enough, enough-qc, enough-qc-qs) under which the refinements are possible.

Source reference. Original reference 0937: sources/stacks-chow/tags.txt, source lines 11505–11505. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0938: sources/stacks-chow/tags.txt, source lines 11506–11506. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0939: sources/stacks-chow/tags.txt, source lines 11507–11507. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 093A: sources/stacks-chow/tags.txt, source lines 11508–11508. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0G1W: sources/stacks-chow/tags.txt, source lines 20036–20036. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *In Situation 0937 assume (0938), (0939), and (093A) hold. Let $\mathcal{O}$ be a sheaf of rings. Assume given a map*

$$\bigoplus\nolimits_{j=1,\ldots,m} j_{V_j!}\mathcal{O}_{V_j} \longrightarrow \bigoplus\nolimits_{i=1,\ldots,n} j_{U_i!}\mathcal{O}_{U_i}$$

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with U_i and V_j in $\mathcal{B}$, and coverings $\{U_{ik} \to U_i\}_{k \in K_i}$ with $U_{ik} \in \mathcal{B}$. Then there exist finite subsets $K_i' \subset K_i$ and a finite set L of $W_l \in \mathcal{B}$ and a commutative diagram

$$\begin{array}{ccc} \bigoplus_{l \in L} j_{W_l!}\mathcal{O}_{W_l} & \longrightarrow & \bigoplus_{i=1,\ldots,n} \bigoplus_{k \in K_i'} j_{U_{ik}!}\mathcal{O}_{U_{ik}} \\ \downarrow & & \downarrow \\ \bigoplus_{j=1,\ldots,m} j_{V_j!}\mathcal{O}_{V_j} & \longrightarrow & \bigoplus_{i=1,\ldots,n} j_{U_i!}\mathcal{O}_{U_i} \end{array}$$

inducing an isomorphism on cokernels of the horizontal maps.

Proof. Since U_i is quasi-compact, we may choose finite subsets $K_i' \subset K_i$ as in Lemma 0G1W. Then since $\bigoplus_{i=1,\ldots,n} \bigoplus_{k \in K_i'} j_{U_{ik}!}\mathcal{O}_{U_{ik}} \to \bigoplus_{i=1,\ldots,n} j_{U_i!}\mathcal{O}_{U_i}$ is surjective, we can find coverings $\{V_{jm} \to V_j\}_{m \in M_j}$ with $V_{jm} \in \mathcal{B}$ such that we can find a commutative diagram

$$\begin{array}{ccc} \bigoplus_{j=1,\ldots,m} \bigoplus_{m \in M_j} j_{V_{jm}!}\mathcal{O}_{V_{jm}} & \longrightarrow & \bigoplus_{i=1,\ldots n} \bigoplus_{k \in K_i'} j_{U_{ik}!}\mathcal{O}_{U_{ik}} \\ \downarrow & & \downarrow \\ \bigoplus_{j=1,\ldots,m} j_{V_j!}\mathcal{O}_{V_j} & \longrightarrow & \bigoplus_{i=1,\ldots,n} j_{U_i!}\mathcal{O}_{U_i} \end{array}$$

Since V_j is quasi-compact, we can choose finite subsets $M_j' \subset M_j$ as in Lemma 0G1W. Set

$$L = \Big(\coprod\nolimits_{i=1,\ldots,n} K_i' \times K_i'\Big) \coprod \Big(\coprod\nolimits_{j=1,\ldots,m} M_j'\Big)$$

and for $l = (k, k') \in K_i' \times K_i' \subset L$ set $W_l = U_{ik} \times_{U_i} U_{ik'}$ and for $l = m \in M_j' \subset L$ set $W_l = V_{jm}$. Since we have the exact sequences of Lemma 0G1W for the families $\{U_{ik} \to U_i\}_{k \in K_i'}$ we conclude that we get a diagram as in the statement of the lemma (details omitted), except that it is not yet clear that $W_l \in \mathcal{B}$. However, since W_l is quasi-compact for all $l \in L$ we do another application of Lemma 0G1W and find finite families of maps $\{W_{lt} \to W_l\}_{t \in T_l}$ with $W_{lt} \in \mathcal{B}$ such that $\bigoplus j_{W_{lt}!}\mathcal{O}_{W_{lt}} \to j_{W_l!}\mathcal{O}_{W_l}$ is surjective. Then we replace L by $\coprod_{l \in L} T_l$ and everything is clear. $\square$